

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398722
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Park; Honghee et al.

Scroll compressor

Abstract

A scroll compressor is provided that may include an orbiting scroll, a non-orbiting scroll, a back pressure chamber assembly, a back pressure passage, and a flow resistance portion. The back pressure passage may provide communication between a compression chamber and a back pressure chamber, and the flow resistance portion may be disposed at a middle portion of the back pressure passage to reduce an amount of refrigerant flowing through the back pressure passage. This may reduce a substantial cross-sectional area of the back pressure passage while improving machining of the back pressure passage, thereby decreasing an amount of refrigerant flowing between the compression chamber and the back pressure chamber. Accordingly, pressure pulsation in the back pressure chamber may be lowered and sealing stability between the orbiting scroll and the non-orbiting scroll may be enhanced, thereby improving compression performance.

Inventors: Park; Honghee (Seoul, KR), Kim; Wooyoung (Seoul, KR), Yu; Byeonghun (Seoul, KR)

Applicant: LG ELECTRONICS INC. (Seoul, KR)

Family ID: 1000008780937

Assignee: LG ELECTRONICS INC. (Seoul, KR)

Appl. No.: 18/588301

Filed: February 27, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240318653 A1	Sep. 26, 2024

Foreign Application Priority Data

KR	10-2023-0038037	Mar. 23, 2023
----	-----------------	---------------

Publication Classification

Int. Cl.: F04C29/12 (20060101); F04C18/02 (20060101); F04C29/00 (20060101)

U.S. Cl.:

CPC F04C18/0215 (20130101); F04C18/0261 (20130101); F04C29/0035 (20130101); F04C29/12 (20130101);

Field of Classification Search

CPC: F04C (29/0035); F04C (18/0215); F04C (18/0261); F04C (18/0246); F04C (18/0253); F04C (29/12)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4506423	12/1984	Nakamura	251/126	F25B 41/30
10527041	12/2019	Moon	N/A	F04C 29/028
2012/0107163	12/2011	Monnier et al.	N/A	N/A
2015/0345493	12/2014	Lochner et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
113931842	12/2021	CN	F04C 18/02
2020-033881	12/2019	JP	N/A
10-2018-0094219	12/2017	KR	N/A
10-2060475	12/2018	KR	N/A

OTHER PUBLICATIONS

English translation of CN 113931842 by PE2E Jun. 21, 2024. cited by examiner
Korean Office Action dated Jul. 22, 2024 issued in Application No. 10-2023-0038037. cited by applicant

Primary Examiner: Wan; Deming

Attorney, Agent or Firm: KED & ASSOCIATES

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) Pursuant to 35 U.S.C. § 119(a), this application claims the benefit of the earlier filing date and the right of priority to Korean Patent Application No. 10-2023-0038037, filed in Korea on Mar. 23,

2023, the contents of which are incorporated by reference herein in their entirety.

BACKGROUND

1. Field

(2) A scroll compressor is disclosed herein.

2. Background

(3) A scroll compressor is configured such that an orbiting scroll and a non-orbiting scroll are engaged with each other and a pair of compression chambers is formed between the orbiting scroll and the non-orbiting scroll while the orbiting scroll performs an orbiting motion with respect to the non-orbiting scroll. In the scroll compressor, as the pair of compression chambers is formed, leakage between the compression chambers may be suppressed or prevented only when the non-orbiting scroll and the orbiting scroll are sealed in close contact in an axial direction. Thus, the scroll compressor employs a back pressure structure which presses the orbiting scroll toward the non-orbiting scroll or presses the non-orbiting scroll toward the orbiting scroll. The former may be defined as an orbiting back pressure type, and the latter may be defined as a non-orbiting back pressure type.

(4) The orbiting back pressure type is applied to a structure in which the non-orbiting scroll is fixed to a main frame. In the orbiting back pressure type, a back pressure chamber is formed between the orbiting scroll and the main frame supporting the orbiting scroll. The non-orbiting back pressure type is applied to a structure in which the non-orbiting scroll is axially movable relative to the main frame. In the non-orbiting back pressure type, a back pressure chamber is formed on a rear surface of the non-orbiting scroll. U.S. Patent Publication No. US 2015/0345493 and U.S. Patent Publication No. US 2012/0107163, which are hereby incorporated by reference, each disclose a non-orbiting back pressure type scroll compressor.

(5) In these non-orbiting back pressure type scroll compressors, as the non-orbiting scroll is pressed toward the orbiting scroll by a pressure in the back pressure chamber, it is advantageous in terms of efficiency of the compressor to maintain a difference between the pressure in the back pressure chamber and a pressure in the compression chamber as constantly as possible. This is especially true in low load operating conditions (or a low pressure ratio operation) in which a suction pressure of the compression chamber is lowered. Thus, a back pressure control device is required to change the pressure in the back pressure chamber in response to the pressure in the compression chamber.

(6) In these related art scroll compressors, however, it is advantageous to lower pulsation pressure of the back pressure chamber and reduce a dead volume by making an area of the back pressure passage that connects the compression chamber and the back pressure chamber as small as possible, but there are limits to reduction in the area of the back pressure passage due to machining characteristics. As a result, an amount of refrigerant flowing into or flowing out of the back pressure chamber may increase. This may cause not only an increase in pulsation pressure of the back pressure chamber, but also an increase in area of the back pressure passage, thereby increasing the dead volume.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:

(2) FIG. 1 is a longitudinal cross-sectional view of an inner structure of a scroll compressor in accordance with an embodiment;

(3) FIG. 2 is an exploded perspective view of a flow resistance portion in FIG. 1 according to an embodiment;

(4) FIG. 3 is an assembled cross-sectional view of the flow resistance portion of FIG. 2;

- (5) FIG. 4 is a cross-sectional view, taken along line “IV-IV” of FIG. 3;
- (6) FIG. 5 is a cross-sectional view, taken along line “V-V” of FIG. 3;
- (7) FIG. 6 is a cross-sectional view illustrating refrigerant that passes through a back pressure passage in FIG. 3;
- (8) FIG. 7 is an exploded perspective view of a flow resistance portion in FIG. 1 according to another embodiment;
- (9) FIG. 8 is an assembled cross-sectional view of the flow resistance portion of FIG. 7;
- (10) FIG. 9 is a cross-sectional view, taken along line “IX-IX” of FIG. 8;
- (11) FIG. 10 is a cross-sectional view, taken along line “X-X” of FIG. 8;
- (12) FIG. 11 is an exploded perspective view of a flow resistance portion in FIG. 1 according to still another embodiment;
- (13) FIG. 12 is an assembled planar view of the flow resistance portion of FIG. 11; and
- (14) FIG. 13 is a cross-sectional view, taken along line “XIII-XIII” of FIG. 12.

DETAILED DESCRIPTION

(15) Description will now be given of a scroll compressor according to exemplary embodiments disclosed herein, with reference to the accompanying drawings. Wherever possible, the same or like reference numerals have been used to indicate the same or like components, and repetitive disclosure has been omitted.

(16) Typically, a scroll compressor may be classified as an open type or a hermetic type depending on whether a drive unit (motor unit) and a compression unit are all installed in an inner space of a casing. The former is a compressor in which the motor unit configuring the drive unit is provided separately from the compression unit, and the latter hermetic type is a compressor in which both the motor unit and the compression unit are disposed inside of the casing. Hereinafter, a hermetic type scroll compressor will be described as an example, but embodiments are not necessarily limited to the hermetic scroll compressor. In other words, the embodiments may be equally applied even to the open type scroll compressor in which the motor unit and the compression unit are disposed separately from each other.

(17) In addition, scroll compressors may be classified into a vertical scroll compressor in which a rotary shaft is disposed perpendicular to the ground and a horizontal (lateral) scroll compressor in which the rotary shaft is disposed parallel to the ground. For example, in the vertical scroll compressor, an upper side may be defined as an opposite side to the ground and a lower side may be defined as a side facing the ground. Hereinafter, the vertical scroll compressor will be described as an example. However, the embodiments may also be equally applied to the horizontal scroll compressor. Hereinafter, it will be understood that an axial direction is an axial direction of the rotary shaft, a radial direction is a radial direction of the rotary shaft, the axial direction is an upward and downward (or vertical) direction, and the radial direction is a leftward and rightward or lateral direction, respectively.

(18) FIG. 1 is a longitudinal cross-sectional view of an inner structure of a scroll compressor in accordance with an embodiment. Referring to FIG. 1, a scroll compressor according to an embodiment may include a drive motor **120** disposed in a lower half portion of a casing **110**, and a main frame **130**, an orbiting scroll **140**, a non-orbiting scroll **150**, and a back pressure chamber assembly **160** that form a compression unit disposed above the drive motor **120**. The motor unit is coupled to one or a first end of a rotary shaft **125**, and the compression unit is coupled to another or a second end of the rotary shaft **125**. Accordingly, the compression unit may be connected to the motor unit by the rotary shaft **125** to be operated by a rotational force of the motor unit.

(19) The casing **110** may include a cylindrical shell **111**, an upper cap **112**, and a lower cap **113**. The cylindrical shell **111** has a cylindrical shape with upper and lower ends open, and the drive motor **120** and the main frame **130** may be fitted on an inner circumferential surface of the cylindrical shell **111**. A terminal bracket (not illustrated) may be coupled to an upper half portion of the cylindrical shell **111**. A terminal (not illustrated) that transmits external power to the drive

motor **120** may be coupled through the terminal bracket. In addition, a refrigerant suction pipe **117** described hereinafter is coupled to the upper portion of the cylindrical shell **111**, for example, above the drive motor **120**.

(20) The upper cap **112** may be coupled to cover the upper open end of the cylindrical shell **111**. The lower cap **113** may be coupled to cover the lower open end of the cylindrical shell **111**. A rim of a high/low pressure separation plate **115** described hereinafter may be inserted between the cylindrical shell **111** and the upper cap **112** to be, for example, welded on the cylindrical shell **111** and the upper cap **112**. A rim of a support bracket **116** described hereinafter may be inserted between the cylindrical shell **111** and the lower cap **113** to be, for example, welded on the cylindrical shell **111** and the lower cap **113**. Accordingly, the inner space of the casing **110** may be sealed.

(21) The rim of the high/low pressure separation plate **115** may be welded on the casing **110** as described above. A central portion of the high/low pressure separation plate **115** may be bent to protrude toward an upper surface of the upper cap **112** so as to be disposed above the back pressure chamber assembly **160** described hereinafter. The refrigerant suction pipe **117** may communicate with a space below the high/low pressure separation plate **115**, and a refrigerant discharge pipe **118** may communicate with a space above the high/low pressure separation plate **115**. Accordingly, a low-pressure part or portion **110a** forming a suction space may be formed below the high/low pressure separation plate **115**, and a high-pressure part or portion **110b** of a discharge space may be formed above the high/low pressure separation plate **115**.

(22) In addition, a through hole **115a** may be formed through a center of the high/low pressure separation plate **115**. A sealing plate **1151** from which a floating plate **165** described hereinafter may be detachable is inserted into the through hole **115a**. The low-pressure portion **110a** and the high-pressure portion **110b** may be blocked from each other by attachment/detachment of the floating plate **165** and the sealing plate **1151** or may communicate with each other through a high/low pressure communication hole **1151a** of the sealing plate **1151**.

(23) In addition, the lower cap **113** may define an oil storage space **110c** together with a lower portion of the cylindrical shell **111** forming the low-pressure portion **110a**. In other words, the oil storage space **110c** may be defined in a lower portion of the low-pressure portion **110a**. The oil storage space **110c** thus defines a part or portion of the low-pressure portion **110a**.

(24) Referring to FIG. 1, the drive motor **120** according to this embodiment may be disposed in a lower half portion of the low-pressure portion **110a** and may include a stator **121** and a rotor **122**. The stator **121** may be, for example, shrink-fitted to an inner wall surface of the cylindrical shell **111**, and the rotor **122** may be rotatably disposed inside of the stator **121**.

(25) The stator **121** includes a stator core **1211** and a stator coil **1212**. The stator core **1211** may be formed in a cylindrical shape and, for example, shrink-fitted onto the inner circumferential surface of the cylindrical shell **111**. The stator coil **1212** may be wound around the stator core **1211** and electrically connected to an external power source through a terminal (not illustrated) that is coupled through the casing **110**.

(26) The rotor **122** may include a rotor core **1221** and permanent magnets **1222**. The rotor core **1221** may be formed in a cylindrical shape, and rotatably inserted into the stator core **1211** with a preset or predetermined gap therebetween. The permanent magnets **1222** may be embedded in the rotor core **1222** at preset or predetermined intervals along a circumferential direction.

(27) In addition, the rotary shaft **125** is press-fitted to a center of the rotor core **1221**. An orbiting scroll **140** described hereinafter may be eccentrically coupled to an upper end of the rotary shaft **125**. Accordingly, the rotational force of the drive motor **120** may be transmitted to the orbiting scroll **140** through the rotary shaft **125**.

(28) An eccentric portion **1251** that is eccentrically coupled to the orbiting scroll **140** described hereinafter may be formed on an upper end of the rotary shaft **125**. An oil pickup **126** that suctions up oil stored in the lower portion of the casing **110** may be disposed in or at a lower end of the

rotary shaft **125**. An oil passage **1252** may be formed through an inside of the rotary shaft **125** in the axial direction.

(29) Referring to FIG. **1**, the main frame **130** may be disposed on an upper side of the drive motor **120**, and may be, for example, shrink-fitted to or welded on an inner wall surface of the cylindrical shell **111**. The main frame **130** may include a main flange portion **131**, a main bearing portion **132**, an orbiting space portion **133**, a scroll support portion **134**, an Oldham ring support portion **135**, and a frame fixing portion **136**.

(30) The main flange portion **131** may be formed in an annular shape and accommodated in the low-pressure portion **110a** of the casing **110**. An outer diameter of the main flange portion **131** may be smaller than an inner diameter of the cylindrical shell **111** so that an outer circumferential surface of the main flange portion **131** is spaced apart from an inner circumferential surface of the cylindrical shell **111**. However, the frame fixing portion **136** described hereinafter may protrude from an outer circumferential surface of the main flange portion **131** in the radial direction. An outer circumferential surface of the frame fixing portion **136** may be fixed in close contact with the inner circumferential surface of the casing **110**. Accordingly, the frame **130** may be fixedly coupled to the casing **110**.

(31) The main bearing portion **132** may protrude downward from a lower surface of a central part or portion of the main flange portion **131** toward the drive motor **120**. A bearing hole **132a** formed in a cylindrical shape may penetrate through the main bearing portion **132** in the axial direction. The rotary shaft **125** may be inserted into an inner circumferential surface of the bearing hole **132a** and supported in the radial direction.

(32) The orbiting space portion **133** may be recessed from the center portion of the main flange portion **131** toward the main bearing portion **132** to have a predetermined depth and outer diameter. The outer diameter of the orbiting space portion **133** is larger than an outer diameter of a rotary shaft coupling portion **143** that is disposed on the orbiting scroll **140** described hereinafter. Accordingly, the rotary shaft coupling portion **143** may be pivotally accommodated in the orbiting space portion **133**.

(33) The scroll support portion **134** may be formed in an annular shape on an upper surface of the main flange portion **131** along a circumference of the orbiting space portion **133**. Accordingly, the scroll support portion **134** may support the lower surface of an orbiting end plate **141** described hereinafter in the axial direction.

(34) The Oldham ring support portion **135** may be formed in an annular shape on an upper surface of the main flange portion **131** along an outer circumferential surface of the scroll support portion **134**. Accordingly, an Oldham ring **170** may be inserted into the Oldham ring supporting portion **135** to be pivotable.

(35) The frame fixing portion **136** may extend radially from an outer circumference of the Oldham ring support portion **135**. The frame fixing portion **136** may extend in an annular shape or extend to form a plurality of protrusions spaced apart from one another by preset or predetermined distances. This embodiment illustrates an example in which the frame fixing portion **136** has a plurality of protrusions along the circumferential direction.

(36) Referring to FIG. **1**, the orbiting scroll **140** according to this embodiment may be coupled to the rotary shaft **125** to be disposed between the main frame **130** and the non-orbiting scroll **150**. The Oldham ring **170**, which is an anti-rotation mechanism, may be disposed between the main frame **130** and the orbiting scroll **140**. Accordingly, the orbiting scroll **140** may perform an orbiting motion relative to the non-orbiting scroll **150** while its rotational motion is restricted.

(37) The orbiting scroll **140** may include an orbiting end plate **141**, an orbiting wrap **142**, and the rotary shaft coupling portion **143**. The orbiting end plate **141** may be formed approximately in a disk shape. An outer diameter of the orbiting end plate **141** may be mounted on the scroll support portion **134** of the main frame **130** to be supported in the axial direction. Accordingly, the orbiting end plate **141** and the scroll support portion **134** facing it defines an axial bearing surface (no

reference numeral given).

(38) The orbiting wrap **142** may be formed in a spiral shape by protruding from an upper surface of the orbiting end plate **141** facing the non-orbiting scroll **150** to a preset or predetermined height. The orbiting wrap **142** may be formed to correspond to the non-orbiting wrap **152** to perform an orbiting motion by being engaged with a non-orbiting wrap **152** of the non-orbiting scroll **150** described hereinafter. The orbiting wrap **142** defines compression chambers V together with the non-orbiting wrap **152**.

(39) The compression chambers V may include first compression chamber V1 and second compression chamber V2 based on the orbiting wrap **142**. Each of the first compression chamber V1 and the second compression chamber V2 includes a suction pressure chamber (not illustrated), an intermediate pressure chamber (not illustrated), and a discharge pressure chamber (not illustrated) that are continuously formed. Hereinafter, description will be given under the assumption that a compression chamber defined between an outer surface of the orbiting wrap **142** and an inner surface of the non-orbiting wrap **152** facing the same is defined as the first compression chamber V1, and a compression chamber defined between an inner surface of the orbiting wrap **142** and an outer surface of the non-orbiting wrap **152** facing the same is defined as the second compression chamber V2.

(40) The rotary shaft coupling portion **143** protrudes from a lower surface of the orbiting end plate **141** toward the main frame **130**. The rotary shaft coupling portion **143** may be formed in a cylindrical shape, so that an orbiting bearing (not illustrated) configured as a bush bearing can be press-fitted.

(41) Referring to FIG. 1, the non-orbiting scroll **150** according to this embodiment may be disposed on or at an upper portion of the main frame **130** with the orbiting scroll **140** interposed therebetween. The non-orbiting scroll **150** may be fixedly coupled to the main frame **130** or may be coupled to the main frame **130** to be movable up and down. This embodiment illustrates an example in which the non-orbiting scroll **150** is coupled to the main frame **130** to be movable relative to the main frame **130** in the axial direction.

(42) Referring to FIG. 1, the non-orbiting scroll **150** according to this embodiment may include a non-orbiting end plate portion **151**, a non-orbiting wrap **152**, a non-orbiting side wall portion **153**, and a guide protrusion **154**. The non-orbiting end plate portion **151** may be formed in a disk shape and disposed in a lateral direction in the low-pressure portion **110a** of the casing **110**. A discharge port **1511**, a bypass hole **1512**, and a scroll back pressure hole **1811** defining a portion of the back pressure passage **181** described hereinafter may be formed through a central portion of the non-orbiting end plate portion **151** in the axial direction.

(43) The single discharge port **1511** may be formed such that discharge pressure chambers (no reference numerals given) of both compression chambers V1 and V2 formed at inner and outer sides of the non-orbiting wrap **152** communicate with each other. However, in some cases, the discharge port **1511** may be provided as a plurality to communicate with the compression chambers V1 and V2 independently.

(44) The bypass holes **1512** may independently communicate with the both compression chambers V2. In other words, the bypass hole **1512** may be formed closer to a suction side than the discharge port **1511**, and be disposed at one position for each compression chamber V1 and V2. However, in some cases, the bypass holes **1512** may be formed at a plurality of positions for each compression chamber V1 and V2 at predetermined distances along a formation direction of the compression chambers V1 and V2. Although three bypass holes are illustrated in the drawing, hereinafter, it will be defined and described as being formed at one position.

(45) The scroll back pressure hole **1811** may be formed at a position spaced apart from the discharge port **1511** and the bypass holes **1512**. In other words, the scroll back pressure hole **1811** may be formed closer to a suction side than the bypass hole **1512**. However, in some cases, a portion of the scroll back pressure hole **1811** may be disposed between the discharge port **1511** and

the bypass hole **1512**. This embodiment shows an example in which the scroll back pressure hole **1811** is formed at a suction side compared to the bypass hole **1512**. Accordingly, the discharge port **1511**, the bypass hole **1512**, and the scroll back pressure hole **1811** may be disposed sequentially from a discharge side to the suction side in the non-orbiting end plate portion **151**.

(46) In addition, the scroll back pressure hole **1811** may be formed to communicate with each of both compression chambers V1 and V2, but may alternatively be formed to communicate with only one compression chamber of the compression chambers V1 and V2. This embodiment shows an example in which the scroll back pressure hole is formed to communicate with only one compression chamber of the compression chambers V1 and V2.

(47) Additionally, a flow resistance part or portion **182** and/or a portion of the flow resistance portion **182** may be inserted into the scroll back pressure hole **1811**. This embodiment shows an example in which the portion of the flow resistance portion **182** is inserted into the scroll back pressure hole **1811**. In other words, the portion of the flow resistance portion **182** may be inserted into the scroll back pressure hole **1811** and a remaining portion of the flow resistance portion **182** may be inserted into the plate back pressure hole **1812**. Accordingly, an amount of refrigerant flowing through the back pressure passage **181** configured by the scroll back pressure hole **1811** and the plate back pressure hole **1812** may be reduced. The flow resistance portion **182** will be described hereinafter together with the plate back pressure hole **1812**.

(48) The non-orbiting wrap **152** may extend from a compression surface of the non-orbiting end plate **151** facing the orbiting scroll **140** by a preset or predetermined height in the axial direction. The non-orbiting wrap **142** may extend to be spirally rolled a plurality of times toward the non-orbiting side wall portion **153** in a vicinity of the discharge port **1511**. The non-orbiting wrap **152** may be formed to correspond to an orbiting wrap **142**, so as to define a pair of compression chambers V with the orbiting wrap **142**.

(49) The non-orbiting side wall portion **153** may extend in an annular shape from a rim of a compression surface of the non-orbiting end plate **151** in the axial direction to surround the non-orbiting wrap **152**. A suction port **1531** may be formed through one side of an outer circumferential surface of the non-orbiting side wall portion **153** in the radial direction.

(50) The guide protrusion **154** may extend radially from an outer circumferential surface of a lower side of the non-orbiting side wall portion **153**. The guide protrusion **154** may be formed in a single annular shape or may be provided as a plurality disposed at preset or predetermined distances in the circumferential direction. This embodiment will be mainly described based on an example in which a plurality of guide protrusions **154** is disposed at preset or predetermined distances along the circumferential direction.

(51) Referring to FIG. **1**, the back pressure chamber assembly **160** according to this embodiment may be disposed at the rear surface of the non-orbiting scroll **150**. Accordingly, a back pressure of the back pressure chamber **160a** (more specifically, a force that the back pressure acts on the back pressure chamber) is applied to the non-orbiting scroll **150**. In other words, the non-orbiting scroll **150** is pressed toward the orbiting scroll **140** by the back pressure to seal both the compression chambers V1 and V2.

(52) The back pressure chamber assembly **160** may include back pressure plate **161** and floating plate **165**. The back pressure plate **161** may be coupled to an upper surface of the non-orbiting end plate **151**. A floating plate **165** may be slidably coupled to the back pressure plate **161** to define the back pressure chamber **160a** together with the back pressure plate **161**. The back pressure plate **161** may include a fixed plate portion **1611**, a first annular wall portion **1612**, and a second annular wall portion **1613**.

(53) A plate back pressure hole **1812**, which communicates with the back pressure chamber **160a**, may be formed through the fixed plate portion **1611** in the axial direction. The plate back pressure hole **1812** may communicate with the compression chamber V through the scroll back pressure hole **1811**. Accordingly, the compression chamber V and the back pressure chamber **160a** may

communicate with each other through the plate back pressure hole **1812** and the scroll back pressure hole **1811**.

(54) The plate back pressure hole **1812** may be formed to correspond to the scroll back pressure hole **1811**. For example, the plate back pressure hole **1812** may communicate with the scroll back pressure hole **1811**. A portion of the flow resistance portion **182** described above may be fixedly inserted into the plate back pressure hole **1812**. Accordingly, the amount of refrigerant flowing from the compression chamber V to the back pressure chamber **160a** may be reduced, thereby reducing not only pressure pulsation in the back pressure chamber **160a** but also a dead volume in the back pressure passage **181** including the plate back pressure hole **1812**. The plate back pressure hole **1812** and the scroll back pressure hole **1811** will be described hereinafter again together with the flow resistance portion **182**.

(55) The first annular wall portion **1612** and the second annular wall portion **1613** may be formed on an upper surface of the fixed plate portion **1611** to surround inner and outer circumferential surfaces of the fixed plate portion **1611**. Accordingly, the back pressure chamber **160a** formed in the annular shape is defined by an outer circumferential surface of the first annular wall portion **1612**, an inner circumferential surface of the second annular wall portion **1613**, the upper surface of the fixed plate portion **1611**, and a lower surface of the floating plate **165**.

(56) The first annular wall portion **1612** may include an intermediate discharge port **1612a** that communicates with the discharge port **1511** of the non-orbiting scroll **150**. A valve guide groove **1612b** into which a discharge valve **155** is slidably inserted may be formed at an inner side of the intermediate discharge port **1612a**. A backflow prevention hole **1612c** may be formed in a center of the valve guide groove **1612b**. Accordingly, the discharge valve **155** may be selectively opened and closed between the discharge port **1511** and the intermediate discharge port **1612a** to suppress or prevent discharged refrigerant from flowing back into the compression chambers V1 and V2.

(57) The floating plate **165** may be formed in an annular shape. The floating plate **165** may be formed of a lighter material than the back pressure plate **161**. Accordingly, the floating plate **165** may be detachably coupled to a lower surface of the high/low pressure separation plate **115** while moving in the axial direction with respect to the back pressure plate **161** depending on the pressure of the back pressure chamber **160a**. For example, when the floating plate **165** is brought into contact with the high/low pressure separation plate **115**, the floating plate **165** serves to seal the low-pressure portion **110a** such that the discharged refrigerant is discharged to the high-pressure portion **110b** without leaking into the low-pressure portion **110a**.

(58) The scroll compressor according to this embodiment may operate as follows.

(59) That is, when power is applied to the drive motor **120** and a rotational force is generated, the orbiting scroll **140** eccentrically coupled to the rotary shaft **125** performs an orbiting motion relative to the non-orbiting scroll **150** due to the Oldham ring **170**. During this process, the first compression chamber V1 and the second compression chamber V2 that continuously move are formed between the orbiting scroll **140** and the non-orbiting scroll **150**. The first compression chamber V1 and the second compression chamber V2 are gradually reduced in volume from the suction port (or suction chamber) **1531** to the discharge port (or discharge chamber) **1511** during the orbiting motion of the orbiting scroll **140**.

(60) Accordingly, refrigerant is suctioned into the low-pressure portion **110a** of the casing **110** through the refrigerant suction pipe **117**. Some of this refrigerant is suctioned directly into the suction pressure chambers (no reference numerals given) of the first compression chamber V1 and the second compression chamber V2, respectively, while the remaining refrigerant first flows toward the drive motor **120** to cool down the drive motor **120** and then is suctioned into the suction pressure chambers (no reference numerals given).

(61) The refrigerant is compressed while moving along moving paths of the first compression chamber V1 and the second compression chamber V2. The compressed refrigerant partially flows into the back pressure chamber **160a** formed by the back pressure plate **161** and the floating plate

165 through the scroll back pressure hole **1811** and the plate back pressure hole **1812** before reaching the discharge port **1511**. Accordingly, the back pressure chamber **160a** forms an intermediate pressure.

(62) The floating plate **165** then rises toward the high/low pressure separation plate **115** to be brought into close contact with the sealing plate **1151** provided on the high/low pressure separation plate **115**. The high-pressure portion **110b** of the casing **110** is separated from the low-pressure portion **110a**, to prevent the refrigerant discharged from each compression chamber V1 and V2 from flowing back into the low-pressure portion **110a**.

(63) On the other hand, the back pressure plate **161** is pressed down toward the non-orbiting scroll **150** by the pressure of the back pressure chamber **160a**. The non-orbiting scroll **150** is pressed toward the orbiting scroll **140**. Accordingly, the non-orbiting scroll **150** may be brought into close contact with the orbiting scroll **140**, thereby preventing the refrigerant inside of both compression chambers from leaking from a high-pressure compression chamber forming an intermediate pressure chamber to a low-pressure compression chamber.

(64) The refrigerant is compressed to a set pressure while moving from the intermediate pressure chamber toward a discharge pressure chamber. This refrigerant moves to the discharge port **1511** and presses the discharge valve **155** in an opening direction. Responsive to this, the discharge valve **155** is pushed up along the valve guide groove **1612b** by the pressure of the discharge pressure chamber, so as to open the discharge port **1511**. The refrigerant in the discharge pressure chamber flows to the high-pressure portion **110b** through the discharge port **1511** and the intermediate discharge port **1612a** disposed in the back pressure plate **161**.

(65) As described above, in the non-orbiting back pressure type scroll compressor, it is advantageous that the cross-sectional area (or inner diameter) of the back pressure passage **181**, which connects the compression chamber V and the back pressure chamber **160a**, is as small as possible, in terms of reducing dead volume in the back pressure passage **181** as well as pressure pulsation in the back pressure chamber **160a**. However, there are limits in the reduction of the cross-sectional area due to machining characteristics. This may unnecessarily increase the cross-sectional area of the back pressure passage **181** and thereby increase the pressure pulsation in the back pressure chamber **160a**, which may cause an increase in friction loss between scrolls or deterioration of compression performance. In addition, as the cross-sectional area of the back pressure passage **181** increases, the dead volume in the back pressure passage **181** may increase, thereby lowering compression efficiency. Accordingly, in this embodiment, the cross-sectional area of the back pressure passage may be minimized by providing the flow resistance portion in the middle of the back pressure passage, thereby lowering the pressure pulsation in the back pressure chamber and reducing the dead volume.

(66) FIG. 2 is an exploded perspective view of a flow resistance portion in FIG. 1 according to an embodiment. FIG. 3 is an assembled cross-sectional view of the flow resistance portion of FIG. 2. FIG. 4 is a cross-sectional view, taken along line "IV-IV" of FIG. 3. FIG. 5 is a cross-sectional view, taken along line "V-V" of FIG. 3, and FIG. 6 is a cross-sectional view illustrating refrigerant that passes through a back pressure passage in FIG. 3.

(67) Referring to FIGS. 2 to 6, in the scroll compressor according to this embodiment, the flow resistance portion **182** may be inserted in the middle of the back pressure passage **181** that connects the compression chamber V and the back pressure chamber **160a**. The flow resistance portion **182** may be fixedly inserted into the back pressure passage **181**. Accordingly, the cross-sectional area of the back pressure passage **181** may be reduced by the cross-sectional area of the flow resistance portion **182**, resulting in a great reduction in the actual cross-sectional area of the back pressure passage **181**, that is, the cross-sectional area of the refrigerant passage **181a** between the inner circumferential surface of the back pressure passage **181** and the outer circumferential surface of the flow resistance portion **182**. With this structure, the inner diameter of the back pressure passage **181** may increase to improve machining of the back pressure passage **181** and decrease the cross-

sectional area of the refrigerant passage **181a** substantially forming the back pressure passage **181**, which may result in reducing the amount of refrigerant flowing through the back pressure passage **181**.

(68) The back pressure passage **181** may include a scroll back pressure hole **1811** and a plate back pressure hole **1812**. The scroll back pressure hole **1811** and the plate back pressure hole **1812** may communicate with each other, and may be formed so that the flow resistance portion **182**, which will be described hereinafter, may be inserted therethrough in the axial direction. Accordingly, the flow resistance portion **182** may be integrally formed, thereby enhancing machining and assembling properties for the flow resistance portion **182**.

(69) Referring to FIGS. 2 and 3, the scroll back pressure hole **1811** according to this embodiment may include a small-diameter portion **1811a** and a large-diameter portion **1811b**. The small-diameter portion **1811a** may be a portion which has one end that communicates with the compression chamber V, and the large-diameter portion **1811b** may be a portion which has a larger inner diameter than the small-diameter portion **1811a** and into which a lower-half portion of the flow resistance portion **182** is inserted. For example, as illustrated in FIG. 4, the small-diameter portion **1811a** and the large-diameter portion **1811b** may be formed on a same axis, but a cross-sectional area of the small-diameter portion **1811a** may be smaller than a cross-sectional area of the flow resistance portion **182**. This may suppress or prevent the flow resistance portion **182** from being pushed toward the compression chamber V even if the pressure of the back pressure chamber **160a** increases more than the pressure of the compression chamber V.

(70) However, in some cases, the small-diameter portion **1811a** and the large-diameter portion **1811b** may be formed on different axes, but may have a same inner diameter. In this case, as a center of the flow resistance portion **182** and a center of the small-diameter portion **1811a** are located on different axes, separation of the flow resistance portion **182** may be suppressed or prevented even if the small-diameter portion **1811a** has a large inner diameter.

(71) As illustrated in FIGS. 3 and 4, the cross-sectional area of the large-diameter portion **1811b** may be larger than the cross-sectional area of the flow resistance portion **182**. In other words, a first refrigerant passage groove **1811c** may be radially recessed into an inner circumferential surface of the large-diameter portion **1811b**, to be spaced apart from the outer circumferential surface of the flow resistance portion **182**. The first refrigerant passage groove **1811c** may communicate with a second refrigerant passage groove **1812a**, which will be described hereinafter, to define the refrigerant passage **181a** between the inner circumferential surface of the back pressure passage **181** and the outer circumferential surface of the flow resistance portion **182**. Accordingly, the refrigerant passage **181a** may communicate with the compression chamber V through the small-diameter portion **1811a**.

(72) A cross-sectional area of the first refrigerant passage groove **1811c** may be smaller than or equal to the cross-sectional area of the small-diameter portion **1811a**. For example, the first refrigerant passage groove **1811c** may be formed approximately in an arcuate shape on each of both sides of the scroll back pressure hole **1811**, and a very narrow radial gap may be defined between an inner circumferential surface of the first refrigerant passage groove **1811c** and an outer circumferential surface of the flow resistance portion **182**. Accordingly, the first refrigerant passage groove **1811c**, which substantially defines the portion of the back pressure passage **181** between the compression chamber V and the back pressure chamber **160a**, may have a very small cross-sectional area, thereby reducing the amount of refrigerant flowing between the compression chamber V and the back pressure chamber **160a**.

(73) Although not shown, the large-diameter portion **1811b** and the flow resistance portion **182** may have a same cross-sectional area, and a refrigerant passage groove (not illustrated) may be formed in the outer circumferential surface of the flow resistance portion **182** through D-cutting.

(74) Referring to FIGS. 2 and 3, the plate back pressure hole **1812** according to this embodiment may be formed on a same axis as the scroll back pressure hole **1811**. The plate back pressure hole

1812 may be formed in the same manner as the large-diameter portion **1811b** of the scroll back pressure hole **1811**. In other words, a second refrigerant passage groove **1812a** may be formed in an inner circumferential surface of the plate back pressure hole **1812**. The second refrigerant passage groove **1812a** may communicate with the first refrigerant passage groove **1811c** formed in the inner circumferential surface of the scroll back pressure hole **1811**, to form the refrigerant passage **181a**. (75) Referring to FIG. 5, the second refrigerant passage groove **1812a** may be formed in the same manner as the first refrigerant passage groove **1811c**. Accordingly, even if the flow resistance portion **182** is press-fitted into the plate back pressure hole **1812**, the second refrigerant passage groove **1812a** may be formed between the outer circumferential surface of the flow resistance portion **182** and the plate back pressure hole **1812**.

(76) The second refrigerant passage groove **1812a** may be recessed laterally in the inner circumferential surface of the plate back pressure hole **1812**, like the first refrigerant passage groove **1811c**, and may be extend lengthwise in the axial direction, for example, by a length of the plate back pressure hole **1812**. Accordingly, the second refrigerant passage groove **1812a** may communicate with the first refrigerant passage groove **1811c** formed between the inner circumferential surface of the scroll back pressure hole **1811** and the outer circumferential surface of the flow resistance portion **182**, thereby smoothing the flow of refrigerant between the compression chamber V and the back pressure chambers **160a**.

(77) A cross-sectional area of the second refrigerant passage groove **1812a** may be formed as small as possible, for example, to be smaller than the cross-sectional area of the flow resistance portion **182**. Accordingly, the cross-sectional area of the refrigerant passage **181a**, which substantially defines the back pressure passage **181** between the compression chamber V and the back pressure chamber **160a**, may be decreased, thereby reducing the amount of refrigerant flowing between the compression chamber V and the back pressure chamber **160a**.

(78) Referring to FIGS. 2 and 3, the flow resistance portion **182** according to this embodiment may be formed in the shape of a pin or rod that extends long and thin in the axial direction. For example, the flow resistance portion **182** may be formed as a long integral body such that one or a first end thereof is inserted into the scroll back pressure hole **1811** while another or a second end is inserted into the plate back pressure hole **1812**. Accordingly, the flow resistance portion **182** may extend across from the scroll back pressure hole **1811** to the plate back pressure hole **1812**.

(79) However, a length of the flow resistance portion **182** may be set such that the flow resistance portion **182** is spaced apart from the small-diameter portion **1811a** of the scroll back pressure hole **1811**, namely, may be shorter than a length of the back pressure passage **181**. Accordingly, a lower end of the flow resistance portion **182** may be spaced apart from the small-diameter portion **1811a**, to thus suppress or prevent the small-diameter portion **1811a** from being blocked even if the cross-sectional area of the flow resistance portion **182** is larger than the cross-sectional area of the small-diameter portion **1811a**.

(80) The cross-sectional area of the flow resistance portion **182** may be constant along the axial direction, but may be larger than the cross-sectional area of the small-diameter portion **1811a** of the scroll back pressure hole **1811** and smaller than the cross-sectional area of the large-diameter portion **1811b**. Accordingly, even if the pressure in the back pressure chamber **160a** increases rapidly, the flow resistance portion **182** may be suppressed or prevented from being separated toward the compression chamber V by being axially supported on a stepped surface between the large-diameter portion **1811b** and the small-diameter portion **1811a**.

(81) Although not shown, a press-fitted protrusion (not illustrated) may be formed on a portion of the flow path resistance portion **182**, for example, an upper portion forming the second end of the flow resistance portion **182**. In other words, the press-fitted protrusion may protrude from an upper-half portion of the outer circumferential surface of the flow resistance portion **182** toward the inner circumferential surface of the plate back pressure hole **1812** in the radial direction. In this case, a length of the press-fitted protrusion may be shorter than the length of the second refrigerant

passage groove **1812a** formed in the inner circumferential surface of the plate back pressure hole **1812**. Accordingly, the press-fitted length of the flow resistance portion **182** may be reduced, which may facilitate the flow resistance portion **182** to be press-fitted into the inner circumferential surface of the back pressure passage **181**.

(82) Referring to FIG. 6, when the flow resistance portion **182** is inserted into the back pressure passage **181** as described above, the refrigerant passage **181a**, which is a gap between the inner circumferential surface of the back pressure passage **181** and the outer circumferential surface of the flow resistance portion **182**, substantially defines the back pressure passage **181**. Accordingly, by appropriately adjusting the gap between the inner circumferential surface of the back pressure passage **181** and the outer circumferential surface of the flow resistance portion **182**, the amount of refrigerant flowing between the compression chamber V and the back pressure chamber **160a** may be appropriately reduced.

(83) Therefore, the amount of refrigerant flowing from the compression chamber V into the back pressure chamber **160a** or flowing out from the back pressure chamber **160a** to the compression chamber V may be reduced, thereby lowering pressure pulsation in the back pressure chamber **160a**. With this structure, the pressure of the back pressure chamber **160a**, that is, a pressure fluctuation range of back pressure against the non-orbiting scroll **150**, may be reduced, and thus, an excessive or insufficient contact between both scrolls **140** and **150** may be suppressed or prevented, thereby improving compression efficiency. In addition, as the flow resistance portion **182** is inserted into the back pressure passage **181**, the actual cross-sectional area of the back pressure passage **181** may be reduced and the dead volume due to the back pressure passage **181** may be decreased, thereby improving compression efficiency.

(84) Although not shown, the flow resistance portion **182** may be inserted into only one of the back pressure holes **1811** and **1812** of the scroll back pressure hole **1811** and the plate back pressure hole **1812** defining the back pressure passage **181**. In this case as well, the flow resistance portion **182** may be press-fitted into the scroll back pressure hole **1811** or the plate back pressure hole **1812**. Also, in this case, the length of the flow resistance portion **182** may be shortened to improve machining and assembling properties.

(85) Hereinafter, description will be given of a flow resistance portion according to another embodiment. That is, in the previous embodiment, the flow resistance portion is formed as an integral body to be press-fitted into the back pressure passage. However, in some cases, the flow resistance portion may be configured as a plurality of separate portions, to be inserted into the scroll back pressure hole and the plate back pressure hole, respectively.

(86) FIG. 7 is an exploded perspective view of a flow resistance portion in FIG. 1 according to another embodiment. FIG. 8 is an assembled cross-sectional view of the flow resistance portion of FIG. 7. FIG. 9 is a cross-sectional view, taken along line “IX-IX” of FIG. 8, and FIG. 10 is a cross-sectional view, taken along line “X-X” of FIG. 8.

(87) Referring to FIGS. 7 to 10, the flow resistance portion **182** according to this embodiment may be fixedly inserted into the back pressure passage **181**, similar to that in the previous embodiment, but may be divided into a first portion of the flow resistance portion **182** inserted into the scroll back pressure hole **1811** and a second portion inserted into the plate back pressure hole **1812**. In other words, the flow resistance portion **182** according to this embodiment may include a first flow resistance portion **1821** and a second flow resistance portion **1822**. The first flow resistance portion **1821** is a portion inserted into the scroll back pressure hole **1811**, and the second flow resistance portion **1822** is a portion inserted into the plate back pressure hole **1812**. Accordingly, machining and/or assembling properties of the back pressure passage **181** and flow resistance portion **182** may be improved. Also, the first passage resistance portion **1821** and the second flow resistance portion **1822** may have different cross-sectional areas to further narrow the refrigerant passage **181a**.

(88) More specifically, the scroll back pressure hole **1811**, similar to that in the previous embodiment, may include small-diameter portion **1811a** and a large-diameter portion **1811b**, but

first refrigerant passage **181a1** may be defined between the inner circumferential surface of the large-diameter portion **1811b** and the outer circumferential surface of the first flow resistance portion **1821** to communicate with the small-diameter portion **1811a**. For example, as illustrated in FIG. 9, the outer circumferential surface of the first flow resistance portion **1821** may be D-cut along the axial direction, to define the first refrigerant passage **181a1** between the outer circumferential surface of the first flow resistance portion **1821** and the inner circumferential surface of the large-diameter portion **1811b** facing it. Accordingly, refrigerant in the compression chamber V may flow toward the back pressure chamber **160a** through the small-diameter portion **1811a** and the first refrigerant passage **181a1**, or refrigerant in the back pressure chamber **160a** may flow toward the compression chamber V through the first refrigerant passage **181a1** and the small-diameter portion **1811a**.

(89) Although not shown, the first refrigerant passage **181a1** may alternatively be defined by decreasing the cross-sectional area of the first flow resistance portion **1821** to be smaller than the cross-sectional area of the large-diameter portion **1811b**.

(90) The cross-sectional area of the first refrigerant passage **181a1** may be smaller than or equal to the cross-sectional area of the small-diameter portion **1811a**. For example, the cross-sectional area of the first refrigerant passage **181a1** may be equal to the cross-sectional area of the small-diameter portion **1811a**. Accordingly, the cross-sectional area of the refrigerant passage **181a**, which substantially defines the back pressure passage **181** between the compression chamber V and the back pressure chamber **160a**, may be decreased, thereby reducing the amount of refrigerant flowing between the compression chamber V and the back pressure chamber **160a**. In addition, flow resistance between the small-diameter portion **1811a** and the first refrigerant passage **181a1** may be minimized to enable a smooth flow of refrigerant.

(91) In addition, a first communication groove **1821a** stepped in the axial direction may be formed in one end of the first flow resistance portion **1821** facing the small-diameter portion **1811a**, and the first communication groove **1821a** may communicate with the first refrigerant passage **181a1**. In other words, the one end of the first flow resistance portion **1821** may be stepped to define the first communication groove **181a1** between the small-diameter portion **1811a** and the first refrigerant passage **181a1**. Accordingly, even if one end of the first flow resistance portion **1821** is in close contact with the stepped surface between the small-diameter portion **1811a** and the large-diameter portion **1811b**, the first flow resistance portion **1821** may not completely block the small-diameter portion **1811a** but the small-diameter portion **1811a** and the first refrigerant passage **181a1** may communicate with each other through the first communication groove **1821a**, enabling the flow of refrigerant between the compression chamber V and the back pressure chamber **160a**.

(92) The plate back pressure hole **1812** may be formed with a single inner diameter, as in the previous embodiment, and a second refrigerant passage **181a2** that communicates with the first refrigerant passage **181a1** may be defined between the inner circumferential surface of the plate back pressure hole **1812** and the outer circumferential surface of the second flow resistance portion **1822**. For example, as illustrated in FIG. 10, the cross-sectional area of the second flow resistance portion **1822** may be smaller than the cross-sectional area of the plate back pressure hole **1812**. Accordingly, the outer circumferential surface of the second flow resistance portion **1822** and the inner circumferential surface of the plate back pressure hole **1812** may be spaced apart by the second refrigerant passage **181a2**, so that the refrigerant in the compression chamber V may flow toward the back pressure chamber **160a** through the second refrigerant passage **181a2** or the refrigerant in the back pressure chamber **160a** may flow toward the compression chamber V through the second refrigerant passage **181a2**.

(93) In this case, the cross-sectional area of the plate back pressure hole **1812** may be larger than the cross-sectional area of the scroll back pressure hole **1811**. Accordingly, the first refrigerant passage **181a1** and the second refrigerant passage **181a2** may smoothly communicate with each other.

(94) Although not shown, the second refrigerant passage **181a2**, which may be formed by D-cutting the outer circumferential surface of the second flow resistance portion **1822**, may extend lengthwise in the axial direction. The cross-sectional area of the second refrigerant passage **181a2** may be smaller than or equal to the cross-sectional area of the small-diameter portion **1811a**. For example, the cross-sectional area of the second refrigerant passage **181a2** may be equal to the cross-sectional area of the small-diameter portion **1811a**. Accordingly, the cross-sectional area of the refrigerant passage groove **181a**, which substantially defines the back pressure passage **181** between the compression chamber V and the back pressure chamber **160a**, may be decreased, thereby reducing the amount of refrigerant flowing between the compression chamber V and the back pressure chamber **160a**. In addition, flow resistance between the first refrigerant passage **181a1** and the second refrigerant passage **181a2** may be minimized to enable a smooth flow of refrigerant.

(95) In addition, a second communication groove **1822a** stepped in the axial direction may be formed in one end of the second flow resistance portion **1822** facing the back pressure chamber **160a**. The second communication groove **1822a** may communicate with the second refrigerant passage **181a2**. In other words, the one end of the second flow resistance portion **1822** facing the first flow resistance portion **1821** may be stepped to define the second communication groove **1822a**. Accordingly, even if the second flow resistance portion **1822** is in close contact with the first flow resistance portion **1821**, the first refrigerant passage **181a1** and the second refrigerant passage **181a2** may not be completely blocked by the second flow resistance portion **1822** but may communicate with each other through the second communication groove **1822a**, to enable the flow of refrigerant between the compression chamber V and the back pressure chamber **160a**.

(96) Also, the second flow resistance portion **1822** may be fixed to the back pressure plate **161** by a fastening member **166**. For example, as illustrated in FIG. 8, an upper end of the second flow resistance portion **1822** may be pressed by a head portion **166a** of the fastening member **166**, by which the back pressure plate **161** is fastened to the non-orbiting scroll **150**, so as to be axially supported. In other words, one or a first end of the second flow resistance portion **1822** may be axially supported by the fastening member **166** and another or a second end by the first flow resistance portion **1821**. Accordingly, the first flow resistance portion **1821** and the second flow resistance portion **1822** may be axially fixed between the stepped surface, which is located between the small-diameter portion **1811a** and the large-diameter portion **1811b**, and the fastening member **166** in the close contact state in the axial direction. This may allow selection of various materials for the flow resistance portion **182** and facilitate the flow resistance portion **182** to be fixed to the back pressure passage **181** while lowering a degree of machining the flow resistance portion **182**.

(97) Although not illustrated, the second flow resistance portion **1822** may alternatively be press-fitted to the inner circumferential surface of the plate back pressure hole **1812**. In this case, the second refrigerant passage **181a2** may be formed by being recessed laterally in the plate back pressure hole **1812**, as in the previous embodiment, or may be D-cut into the outer circumferential surface of the second flow resistance portion **1822**.

(98) Referring to FIG. 8, the flow resistance portion **182** may be inserted into the back pressure passage **181** as described above, and when the flow resistance portion **182** is divided into the first flow resistance portion **1821** and the second flow resistance portion **1822**, the machining and assembling properties for the flow resistance portion **182** may be improved. For example, in this embodiment, the first flow resistance portion **1821** and the second flow resistance portion **1822** may be assembled after being separately machined. As the second flow resistance portion **1822** serves as a type of stopper, the back pressure passage **181** may be smoothly formed even if the degree of machining the first flow resistance portion **1821** is lowered.

(99) Additionally, in this embodiment, the cross-sectional area of the first refrigerant passage **181a1** and the cross-sectional area of the second refrigerant passage **181a2** may be different from each other. For example, the first refrigerant passage **181a1** between the outer circumferential surface of

the first flow resistance portion **1821** and the inner circumferential surface of the scroll back pressure hole **1811** may be narrower than the second refrigerant passage **181a2** between the outer circumferential surface of the second flow resistance portion **1822** and the inner circumferential surface of the plate back pressure hole **1812**. Accordingly, the substantial cross-sectional area of the back pressure passage **181** may be reduced to lower the flow amount of refrigerant.

(100) Additionally, in this embodiment, the first flow resistance portion **1821** and the second flow resistance portion **1822** may be formed of different materials. For example, the first flow resistance portion **1821** may be made of a material which is more rigid than a material of the second flow resistance portion **1822**. This may improve the machining for the flow resistance portion **182** while reducing the cost for the flow resistance portion **182**.

(101) Although not shown, in this case as well, the flow resistance portion **182** may be inserted into only one of the back pressure holes **1811** and **1812** of the scroll back pressure hole **1811** or the plate back pressure hole **1812** defining the back pressure passage **181**. In this case as well, the flow resistance portion **182** may be inserted into the plate back pressure hole **1812** to be axially supported by the fastening member **166**, which is fastened to the back pressure plate **161**, or may be inserted into the scroll back pressure hole **1811** to be axially supported by the back pressure plate **161**. Also in this case, the length of the flow resistance portion **182** may be shortened, thereby improving machining and assembling properties.

(102) Hereinafter, description will be given of a flow resistance portion according to still another embodiment. That is, in the previous embodiments, the flow resistance portion is inserted into the scroll back pressure hole and/or the plate back pressure hole forming the back pressure passage **181**, but in some cases, the flow resistance portion may not be inserted into the scroll back pressure hole and/or the plate back pressure hole but may be formed between the scroll back pressure hole and the plate back pressure hole.

(103) FIG. **11** is an exploded perspective view of a flow resistance portion in FIG. **1** according to still another embodiment. FIG. **12** is an assembled planar view of the flow resistance portion of FIG. **11**, and FIG. **13** is a cross-sectional view, taken along line “XIII-XIII” of FIG. **12**.

(104) Referring back to FIG. **11**, the basic configuration of the scroll compressor according to this embodiment and operating effects thereof are similar to those in the previous embodiments. In other words, in the scroll compressor, the internal space of casing **110** is separated into low-pressure portion **110a** and high-pressure portion **110b** by high/low pressure separation plate **115**, drive motor **120** is disposed in the low-pressure portion **110a** of the casing **110**, orbiting scroll **140** is coupled to rotary shaft **125** that transmits a rotational force of the drive motor **120**, and the orbiting scroll **140** forms compression chamber V with non-orbiting scroll **150** while performing orbiting motion relative to the non-orbiting scroll **150**.

(105) In addition, back pressure chamber assembly **160** forming back pressure chamber **160a** is coupled to a rear surface of the non-orbiting scroll **150**, and back pressure passage **181** is formed between the compression chamber V and the back pressure chamber **160a**. Refrigerant in the compression chamber V flows to the back pressure chamber **160a** or refrigerant in the back pressure chamber **160a** flows to the compression chamber **150**, to generate back pressure that presses the non-orbiting scroll **150** toward the orbiting scroll **140**. Accordingly, the scroll compressor may be implemented as the non-orbiting back pressure type, so that the back pressure chamber **160a** may be formed as close to the discharge port **1511** as possible, which enables effective suppression or prevention of refrigerant leakage between the compression chambers, thereby improving compression efficiency.

(106) However, as explained in the previous embodiment, it is advantageous to lower pressure pulsation of the back pressure chamber **160a** and reduce dead volume by forming the cross-sectional area of the back pressure passage **181** as small as possible, but there are limits in reducing the cross-sectional area due to the machining characteristics of the back pressure passage **181**. Accordingly, in this embodiment, the substantial cross-sectional area of the back pressure passage

181 may be reduced without inserting a separate flow resistance portion **186** into the back pressure chamber **181** while the pressure pulsation in the back pressure chamber **160a** may be reduced by lowering the pressure of refrigerant flowing from the compression chamber V to the back pressure chamber **160a**.

(107) Referring to FIGS. **11** to **13**, the flow resistance portion **186** according to this embodiment may be formed in the rear surface of the non-orbiting scroll **150**, the rear surface of the back pressure plate **161** facing the rear surface of the non-orbiting scroll **150**, or a gasket **185** disposed between the non-orbiting scroll **150** and the back pressure plate **161**. In other words, the gasket **185** may be disposed between the rear surface of the non-orbiting scroll **150** and the back pressure plate **161** to seal the discharge port **1511**, the bypass hole **1512**, and the back pressure passage **181**, and the flow resistance portion **186** may be formed in the rear surface of the non-orbiting scroll **150**, the rear surface of the back pressure plate **161**, and/or the gasket **185**. Accordingly, in this embodiment, as the flow resistance portion **186** defines the portion of the back pressure passage **181**, it may be understood that the flow resistance portion **186** is disposed in or at a middle of the back pressure passage **181**. This embodiment illustrates an example in which the flow resistance portion **186** is disposed in the gasket **185**.

(108) For example, the scroll back pressure hole **1811** may be formed in the non-orbiting scroll **150**, the plate back pressure hole **1812** may be formed in the back pressure plate **161**, and the flow resistance portion **186** that connects the scroll back pressure hole **1811** and the plate back pressure hole **1812** may be formed in the gasket **185**. In other words, the scroll back pressure hole **1811** and the plate back pressure hole **1812** may not communicate directly, but may communicate with each other through the flow resistance portion **186** disposed in the gasket **185**. Accordingly, as the scroll back pressure hole **1811** and the plate back pressure hole **1812** are formed on different axes, positions of the scroll back pressure hole **1811** and the plate back pressure hole **1812** may be freely adjusted as needed.

(109) The flow resistance portion **186** may be formed through cutting or by being recessed by a preset or predetermined depth between both sides of the gasket **185**. This embodiment illustrates an example in which the flow resistance portion **186** is recessed by a preset or predetermined depth so that its axial height H is shorter than its lateral width D. Accordingly, a cross-sectional area of the flow resistance portion **186** may be made as small as possible to increase flow resistance as much as possible.

(110) Additionally, the flow resistance portion **186** may extend along the lateral direction to be formed in a linear shape or a curved shape. This embodiment illustrates an example in which the flow resistance portion **186** is formed in a curved shape. Therefore, the flow resistance portion **186** may be formed as long as possible, thereby increasing flow resistance.

(111) As described above, when the flow resistance portion **186** is formed in the gasket **185** or in the rear surface of the non-orbiting scroll **150** or the rear surface of the back pressure plate **161** facing the gasket **185**, the flow resistance portion **186** may be formed even without any separate member, enabling reduction of manufacturing costs. In addition, as the flow resistance portion **186** is formed between the scroll back pressure hole **1811** and the plate back pressure hole **1812**, the flow resistance portion **186** may be made as long as possible to increase flow resistance while allowing the scroll back pressure hole **1811** and the plate back pressure hole **1812** to be formed freely on different axes as needed.

(112) In addition, in the previous embodiments, the structure in which the back pressure chamber assembly **160** including the back pressure plate **161** and the floating plate **165** is separately fastened to the rear surface of the non-orbiting scroll **150** has been described, but the embodiments illustrated in FIGS. **2** and **7** may be equally applied even to a case in which the back pressure plate **161** is omitted and the first annular wall portion **1612** and the second annular wall portion **1613** extend as an integral body from the rear surface of the non-orbiting scroll **150**.

(113) In addition, as described above, the embodiments may be applied equally to open scroll

compressors as well as closed hermetic scroll compressors, applied equally to high-pressure scroll compressors as well as low-pressure scroll compressors, and also applied equally to horizontal scroll compressors as well as vertical scroll compressors.

(114) Embodiments disclosed herein provide a scroll compressor that is capable of lowering pressure pulsation in a back pressure chamber in a non-orbiting back pressure type scroll compressor.

(115) Embodiments disclosed herein further provide a scroll compressor that is capable of reducing an amount of refrigerant flowing between a compression chamber and a back pressure chamber in a non-orbiting back pressure type scroll compressor.

(116) Embodiments disclosed herein furthermore provide a scroll compressor that is capable of facilitating formation of a back pressure passage while reducing a flow amount of refrigerant by forming the back pressure passage to be narrow and long in a non-orbiting back pressure type scroll compressor.

(117) Embodiments disclosed herein also provide a scroll compressor that is capable of reducing a dead volume between a compression chamber and a back pressure chamber in a non-orbiting back pressure type scroll compressor.

(118) Embodiments disclosed herein provide a scroll compressor that may include a casing, an orbiting scroll, a non-orbiting scroll, a back pressure chamber assembly, a back pressure passage, and a flow resistance part or portion. The casing may have a low-pressure part or portion and a high-pressure part or portion. The orbiting scroll may be coupled to a rotary shaft in the low-pressure part of the casing to perform an orbiting motion. The non-orbiting scroll may be engaged with the orbiting scroll to form a compression chamber, and may be movable relative to the orbiting scroll in an axial direction. The back pressure chamber assembly may be disposed on a rear surface of the non-orbiting scroll to form a back pressure chamber. The back pressure passage may be formed such that the compression chamber and the back pressure chamber communicate with each other. The flow resistance part may be disposed in or at a middle portion of the back pressure passage to reduce an amount of refrigerant flowing through the back pressure passage. This may reduce a substantial cross-sectional area of the back pressure passage while improving machining of the back pressure passage, thereby decreasing an amount of refrigerant flowing between the compression chamber and the back pressure chamber. Accordingly, pressure pulsation in the back pressure chamber may be lowered and sealing stability between the orbiting scroll and the non-orbiting scroll may be enhanced, thereby improving compression performance.

(119) The flow resistance part may be inserted into the back pressure passage, and a refrigerant passage may be defined between an inner circumferential surface of the back pressure passage and an outer circumferential surface of the flow resistance portion. The refrigerant passage may have a cross-sectional area smaller than a cross-sectional area of the flow resistance portion. With this structure, the cross-sectional area of the refrigerant passage substantially defining the back pressure passage may be reduced, thereby decreasing an amount of refrigerant flowing through the back pressure passage.

(120) The back pressure passage may include a scroll back pressure hole disposed in the non-orbiting scroll so that one end thereof communicates with the compression chamber, and a plate back pressure hole disposed in the back pressure chamber assembly so that one or a first end thereof communicates with the scroll back pressure hole and another or a second end thereof communicates with the back pressure chamber. The flow resistance part may be inserted across between the scroll back pressure hole and the plate back pressure hole. Accordingly, the flow resistance part may be formed as an integral body, thereby improving machining and assembling properties of the flow resistance part.

(121) For example, the scroll back pressure hole may include a small-diameter portion having one end that communicates with the compression chamber, and a large-diameter portion that extends from another end of the small-diameter portion toward the plate back pressure hole. One or a first

end of the flow resistance portion may be inserted into the large-diameter portion, and another or a second end of the flow resistance part may be inserted into the plate back pressure hole. The flow resistance part may have a cross-sectional area that is smaller than a cross-sectional area of the large-diameter portion and larger than a cross-sectional area of the small-diameter portion. With this structure, the cross-sectional area of the refrigerant passage substantially defining the back pressure passage may be reduced, thereby decreasing an amount of refrigerant flowing through the back pressure passage.

(122) More specifically, a refrigerant passage may be defined between an outer circumferential surface of the flow resistance part and an inner circumferential surface of the large-diameter portion. The refrigerant passage may have a cross-sectional area smaller than or equal to a cross-sectional area of the small-diameter portion. With this structure, an amount of refrigerant flowing through the back pressure passage may be decreased by reducing the cross-sectional area of the refrigerant passage, thereby suppressing or preventing pressure fluctuation (pulsation pressure) occurring in the back pressure chamber.

(123) The flow resistance part may be press-fitted to the back pressure passage, and a refrigerant passage groove may be formed to be recessed laterally into an inner circumferential surface of the back pressure passage to be spaced apart from the outer circumferential surface of the flow resistance part. The refrigerant passage groove may have a cross-sectional area that is smaller than a cross-sectional area of the flow resistance part. With this structure, the cross-sectional area of the refrigerant passage substantially defining the back pressure passage between the compression chamber and the back pressure chamber may be reduced, thereby decreasing an amount of refrigerant flowing between the compression chamber and the back pressure chamber.

(124) The back pressure passage may include a scroll back pressure hole disposed in the non-orbiting scroll so that one end thereof communicates with the compression chamber, and a plate back pressure hole disposed in the back pressure chamber assembly so that one or a first end thereof communicates with the scroll back pressure hole and another or a second end communicates with the back pressure chamber. The flow resistance part may include a first flow resistance portion inserted into the scroll back pressure hole, and a second flow resistance portion disposed at one side of the first flow resistance portion in the axial direction at a spacing from the first flow resistance portion, and inserted into the plate back pressure hole. This may improve machining and assembling properties of the back pressure passage and the flow resistance part while the first flow resistance portion and the second flow resistance portion may have different cross-sectional areas to make the refrigerant passage much narrow.

(125) One or a first end of the second flow resistance portion may be disposed to face one end of the first flow resistance portion, and another or a second end of the second flow resistance portion may be fixed to the back pressure chamber assembly. This may facilitate the flow resistance part to be fixed to the back pressure passage.

(126) More specifically, the scroll back pressure hole may include a small-diameter portion having one end that communicates with the compression chamber, and a large-diameter portion that extends from another end of the small-diameter portion toward the plate back pressure hole. A first refrigerant passage may be defined between an outer circumferential surface of the first flow resistance portion and an inner circumferential surface of the large-diameter portion. The first refrigerant passage may have a cross-sectional area smaller than or equal to a cross-sectional area of the small-diameter portion. With this structure, flow resistance between the small-diameter portion and the first refrigerant passage may be minimized to enable a smooth flow of refrigerant.

(127) More specifically, a first communication groove may be formed stepwise in the axial direction in one end of the first flow resistance portion facing the small-diameter portion, and the first communication groove may communicate with the first refrigerant passage. With this structure, even if one end of the first flow resistance portion is in close contact with a stepped surface between the small-diameter portion and the large-diameter portion, the first flow resistance

portion does not completely block the small-diameter portion so that refrigerant may smoothly flow between the compression chamber and the back pressure chamber.

(128) Also, a second refrigerant passage may be defined between an outer circumferential surface of the second flow resistance portion and an inner circumferential surface of the plate back pressure hole. The second refrigerant passage may have a cross-sectional area smaller than or equal to the cross-sectional area of the small-diameter portion. With this structure, the cross-sectional area of the second refrigerant passage may be minimized to reduce an amount of refrigerant flowing through the back pressure passage.

(129) More specifically, a second communication groove may be formed stepwise in the axial direction in one end of the second flow resistance portion facing the first flow resistance portion, and the first refrigerant passage and the second refrigerant passage may communicate with the second communication groove. Accordingly, even if the second flow resistance portion is in close contact with the first flow resistance portion, the first refrigerant passage and the second refrigerant passage may communicate with each other through the second communication groove, so that refrigerant may smoothly flow between the compression chamber and the back pressure chamber. In addition, flow resistance between the small-diameter portion and the first refrigerant passage may be minimized to enable a smooth flow of refrigerant.

(130) Also, a cross-sectional area of the plate back pressure hole may be larger than a cross-sectional area of the scroll back pressure hole. Accordingly, the first refrigerant passage and the second refrigerant passage may smoothly communicate with each other.

(131) Additionally, the second flow resistance portion may be supported in the axial direction by a fastening member fastened to the back pressure chamber assembly. This may allow selection of various materials for the flow resistance part, and facilitate the flow resistance part to be fixed to the back pressure passage while lowering a degree of machining the flow resistance part.

(132) A gasket may be disposed between the non-orbiting scroll and the back pressure chamber assembly, and the flow resistance part may extend in a lateral direction from at least one of the non-orbiting scroll, the back pressure chamber assembly, and the gasket. With this structure, the flow resistance part may be formed without an addition of a separate member, thereby reducing manufacturing costs.

(133) The back pressure passage may include a scroll back pressure hole disposed in the non-orbiting scroll so that one end thereof communicates with the compression chamber, and a plate back pressure hole disposed in the back pressure chamber assembly so that one or a first end thereof communicates with the scroll back pressure hole and another or a second end communicates with the back pressure chamber. One or a first end of the flow resistance part may be connected to the scroll back pressure hole, and another or a second end of the flow resistance part may be connected to the plate back pressure hole. Both ends of the flow resistance part may be formed on different axes. This may allow positions of the scroll back pressure hole and the plate back pressure hole to be freely adjusted as needed.

(134) More specifically, the flow resistance part may be formed so that an axial height is smaller than a lateral width. Accordingly, the cross-sectional area of the flow resistance part may be made as small as possible to maximize flow resistance.

(135) It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(136) It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region,

layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

(137) Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(138) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(139) Embodiments are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures). As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

(140) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(141) Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

(142) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A scroll compressor, comprising: a casing having a low-pressure portion and a high-pressure portion; an orbiting scroll coupled to a rotary shaft in the low-pressure portion of the casing to perform an orbiting motion; a non-orbiting scroll engaged with the orbiting scroll to form a compression chamber and movable relative to the orbiting scroll in an axial direction; a back pressure chamber assembly disposed on a rear surface of the non-orbiting scroll to form a back pressure chamber; a back pressure passage defined such that the compression chamber and the back pressure chamber communicate with each other; and a flow resistance portion disposed at a middle portion of the back pressure passage to reduce an amount of refrigerant flowing through the back pressure passage, wherein the flow resistance portion is press-fitted into the back pressure passage, wherein at least one refrigerant passage groove is recessed laterally into an inner circumferential surface of the back pressure passage to be spaced apart from an outer circumferential surface of the flow resistance portion, and wherein the at least one refrigerant passage groove has a cross-sectional area that is smaller than a cross-sectional area of the flow resistance portion.
2. The scroll compressor of claim 1, wherein a refrigerant passage is defined between the inner circumferential surface of the back pressure passage and the outer circumferential surface of the flow resistance portion, and wherein the refrigerant passage has a cross-sectional area smaller than the cross-sectional area of the flow resistance portion.
3. The scroll compressor of claim 1, wherein the back pressure passage comprises: a scroll back pressure hole disposed in the non-orbiting scroll so that one end thereof communicates with the compression chamber; and a plate back pressure hole disposed in the back pressure chamber assembly so that a first end thereof communicates with the scroll back pressure hole and a second end communicates with the back pressure chamber, and wherein the flow resistance portion is inserted across between the scroll back pressure hole and the plate back pressure hole.
4. The scroll compressor of claim 3, wherein the scroll back pressure hole comprises: a small-diameter portion having a first end that communicates with the compression chamber; and a large-diameter portion that extends from a second end of the small-diameter portion toward the plate back pressure hole, wherein a first end of the flow resistance portion is inserted into the large-diameter portion and a second end of the flow resistance portion is inserted into the plate back pressure hole, and wherein the cross-sectional area of the flow resistance portion is smaller than a cross-sectional area of the large-diameter portion and larger than a cross-sectional area of the small-diameter portion.
5. The scroll compressor of claim 4, wherein a refrigerant passage is defined between the outer circumferential surface of the flow resistance portion and an inner circumferential surface of the large-diameter portion, and wherein the refrigerant passage has a cross-sectional area smaller than or equal to the cross-sectional area of the small-diameter portion.
6. A scroll compressor, comprising: a casing having a low-pressure portion and a high-pressure portion; an orbiting scroll coupled to a rotary shaft in the low-pressure portion of the casing to perform an orbiting motion; a non-orbiting scroll engaged with the orbiting scroll to form a compression chamber and movable relative to the orbiting scroll in an axial direction; a back pressure chamber assembly disposed on a rear surface of the non-orbiting scroll to form a back pressure chamber; a back pressure passage defined such that the compression chamber and the back pressure chamber communicate with each other; and a flow resistance portion disposed at a middle portion of the back pressure passage to reduce an amount of refrigerant flowing through the back pressure passage, wherein the back pressure passage comprises: a scroll back pressure hole disposed in the non-orbiting scroll so that one end thereof communicates with the compression chamber; and a plate back pressure hole disposed in the back pressure chamber assembly so that a first end thereof communicates with the scroll back pressure hole and a second end communicates with the back pressure chamber, and wherein the flow resistance portion comprises: a first flow resistance portion inserted into the scroll back pressure hole; and a second flow resistance portion

disposed at one side of the first flow resistance portion in the axial direction at a spacing from the first flow resistance portion, and inserted into the plate back pressure hole, wherein the scroll back pressure hole comprises: a small-diameter portion having a first end that communicates with the compression chamber; and a large-diameter portion that extends from a second end of the small-diameter portion toward the plate back pressure hole, wherein a first refrigerant passage is defined between an outer circumferential surface of the first flow resistance portion and an inner circumferential surface of the large-diameter portion, wherein the first refrigerant passage has a cross-sectional area smaller than or equal to a cross-sectional area of the small-diameter portion, wherein a second refrigerant passage is defined between an outer circumferential surface of the second flow resistance portion and an inner circumferential surface of the plate back pressure hole, and wherein the second refrigerant passage has a cross-sectional area smaller than or equal to the cross-sectional area of the small-diameter portion.

7. The scroll compressor of claim 6, wherein a first end of the second flow resistance portion is disposed to face one end of the first flow resistance portion, and a second end of the second flow resistance portion is fixed to the back pressure chamber assembly.

8. The scroll compressor of claim 6, wherein a communication groove is formed stepwise in an axial direction in one end of the first flow resistance portion facing the small-diameter portion, and wherein the communication groove communicates with the first refrigerant passage.

9. The scroll compressor of claim 6, wherein a communication groove is formed stepwise in the axial direction in one end of the second flow resistance portion facing the first flow resistance portion, and wherein the first refrigerant passage and the second refrigerant passage communicate with the communication groove.

10. The scroll compressor of claim 9, wherein the second flow resistance portion is supported in the axial direction by a fastening member fastened to the back pressure chamber assembly.

11. A scroll compressor, comprising: a casing having a low-pressure portion and a high-pressure portion; an orbiting scroll coupled to a rotary shaft in the low-pressure portion of the casing to perform an orbiting motion; a non-orbiting scroll engaged with the orbiting scroll to form a compression chamber and movable relative to the orbiting scroll in an axial direction; a back pressure chamber assembly disposed on a rear surface of the non-orbiting scroll to form a back pressure chamber; a back pressure passage defined such that the compression chamber and the back pressure chamber communicate with each other; and a flow resistance portion disposed at a middle portion of the back pressure passage to reduce an amount of refrigerant flowing through the back pressure passage, wherein a gasket is disposed between the non-orbiting scroll and the back pressure chamber assembly, and wherein the flow resistance portion extends in a lateral direction from at least one of the non-orbiting scroll, the back pressure chamber assembly, or the gasket, wherein the back pressure passage comprises: a scroll back pressure hole disposed in the non-orbiting scroll so that one end thereof communicates with the compression chamber; and a plate back pressure hole disposed in the back pressure chamber assembly so that a first end thereof communicates with the scroll back pressure hole and a second end communicates with the back pressure chamber, wherein a first end of the flow resistance portion is connected to the scroll back pressure hole and a second end of the flow resistance portion is connected to the plate back pressure hole, and wherein ends of the flow resistance portion are on different axes, so that an axial height is smaller than a lateral width.

12. A scroll compressor, comprising: a casing having a low-pressure portion and a high-pressure portion; an orbiting scroll coupled to a rotary shaft in the low-pressure portion of the casing to perform an orbiting motion; a non-orbiting scroll engaged with the orbiting scroll to form a compression chamber and movable relative to the orbiting scroll in an axial direction; a back pressure chamber assembly disposed on a rear surface of the non-orbiting scroll to form a back pressure chamber; a back pressure passage defined such that the compression chamber and the back pressure chamber communicate with each other; and a flow resistance portion disposed at a middle

portion of the back pressure passage to reduce an amount of refrigerant flowing through the back pressure passage, wherein the flow resistance portion is press-fitted into the back pressure passage, wherein at least one refrigerant passage groove is recessed laterally into an inner circumferential surface of the back pressure passage to be spaced apart from an outer circumferential surface of the flow resistance portion.

13. The scroll compressor of claim 12, wherein the flow resistance portion is inserted into the back pressure passage, wherein a refrigerant passage is defined between the inner circumferential surface of the back pressure passage and the outer circumferential surface of the flow resistance portion, and wherein the refrigerant passage has a cross-sectional area smaller than the cross-sectional area of the flow resistance portion.

14. The scroll compressor of claim 12, wherein the back pressure passage comprises: a scroll back pressure hole disposed in the non-orbiting scroll so that one end thereof communicates with the compression chamber; and a plate back pressure hole disposed in the back pressure chamber assembly so that a first end thereof communicates with the scroll back pressure hole and a second end communicates with the back pressure chamber, and wherein the flow resistance portion is inserted across between the scroll back pressure hole and the plate back pressure hole.
